Project ESCAPE: Does Entropy-Based Patching Find Syntax?

Final viva — Team Trimax

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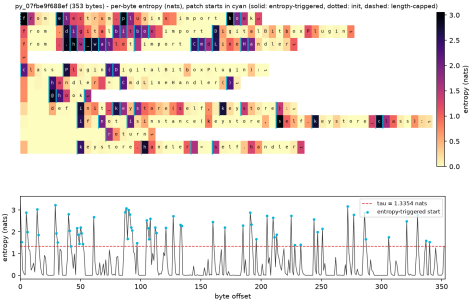
ANLP, October 2026

The question

- BLT starts a new patch when its byte-level entropy model is uncertain: $H_i > \tau$.
- **Grammatical Branching Hypothesis:** uncertainty peaks where a constituent opens, so patches should start at AST / parse constituent starts.
- Tested on 16,300 files (Python, C++, English), held-out split 13,040, against
  - a **density-matched null**: each file's own patch lengths, shuffled, final patch fixed (10,000 permutations);
  - **structure-blind baselines**: newline, indentation change, word start.
- Five objectives (O1–O5), four predictions (P1–P4), plus an extension (a distilled patcher).

Stream A: a faithful patcher, then a distilled one

- HF port `itazap/blt-1b-hf` omits BLT's 512-token sliding window; **we restored it** (8,192-token chunks), float32, $\tau = 1.3354$.
- Without it (HF full attention) the test margin collapses: $+0.023$ vs. $+0.120$ (Py), $+0.008$ vs. $+0.131$ (C++ <8 KB).
- 8 fidelity gates; logits match an independent reimplementation to 1.05×10^{-4}; amended gate set passed on Ada.
- Ada artefacts regenerated; **per-file, score and diagnostic CSVs byte-identical** to the frozen run.
- Extension: 14.3M student distilled from the 99.5M patcher; golden gate reproduced within 0.0000 F1 on GPU.



Stream C: frozen protocol

- One-to-one matching within k bytes; $k \in \{0, 1, 2, 4, 8\}$ chosen on the **calibration split only** $\rightarrow k = 0$ everywhere.
- Fixed-EOF null keeps patch-length multiset and boundary count; paired file bootstrap (2,000).
- Test split scored once; every later analysis reuses the frozen per-file counts or the **same seeded null draws**.
- New checks: repository-clustered bootstrap (Python, 315 repos) widens the CI 3.7$\times$ to [0.112, 0.127]; parser error recovery (28.0% of C++) changes the margin by $+0.004$ [-0.005, 0.015]; files $>$8,191 bytes by -0.002 [-0.008, 0.004].

P1: real alignment, but not beyond line/word segmentation

	F1	H_0 F1	Δ [95% CI]	F1/H_0	vs. best baseline
Python	0.152	0.032	+0.120 [0.117, 0.122]	4.7×	-0.242 (indentation)
C++	0.165	0.036	+0.129 [0.125, 0.134]	4.6×	-0.173 (word-start)
English	0.495	0.147	+0.348 [0.341, 0.354]	3.4×	-0.274 (word-start)

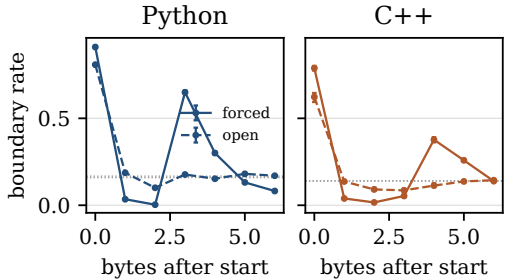
- P1 **supported** against the null ($p \leq 10^{-4}$ everywhere); against the best structure-blind baseline BLT underperforms it in all three languages.
- Precision $\ll$ recall: every boundary has $\geq e^{\tau} \approx 3.8$ live continuations; identifier interiors are flat but not syntactic.
- P2 **supported**: end margins -0.017, -0.035, -0.079.

P3 reversed — because margins scale with target density

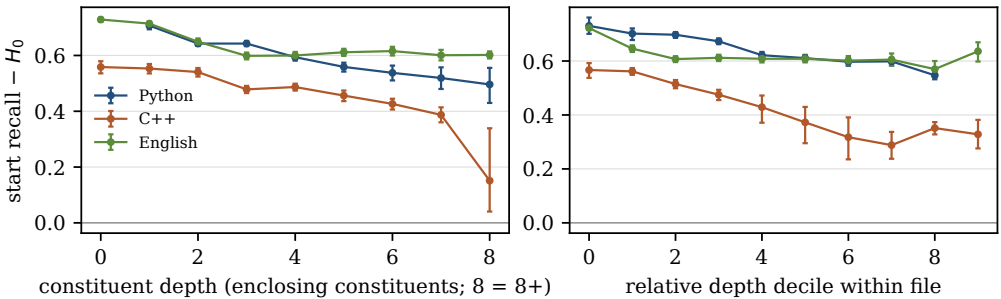
- Predicted: prose near chance. Observed: prose margin $+0.348 >$ code $+0.129$.
- Benepar constituents begin at words; the word-start rule alone gets 0.769 F1 in English.
- On the scale-free ratio to the null, **code aligns more strongly**: 4.7× (Py), 4.6× (C++) vs. 3.4× (En).
- BPP variance half of P3 holds: within-file variance 527.8 (C++), 69.4 (Py) vs. 10.3 (En).

P4 reversed at the start byte — confirmed at the forced bytes

- Start recall margin: forced openers +0.759 vs. open +0.611 (Py) — opposite of P4.
- Test: boundary rate at bytes *after s*, same null draws (start rows reproduce the frozen recalls).
- Rest of keyword: -0.130 (Py), -0.088 (C++) vs. null.
- Byte after keyword: rate 0.000 (Py), 0.035 (C++) vs. null ≈0.159; open-ended stays at the null.
- Boundary returns where choice reopens: Python condition start (rate 0.895); C++ after the forced ((rate 0.377).
- **Forcing is real; the start byte is just before it.**

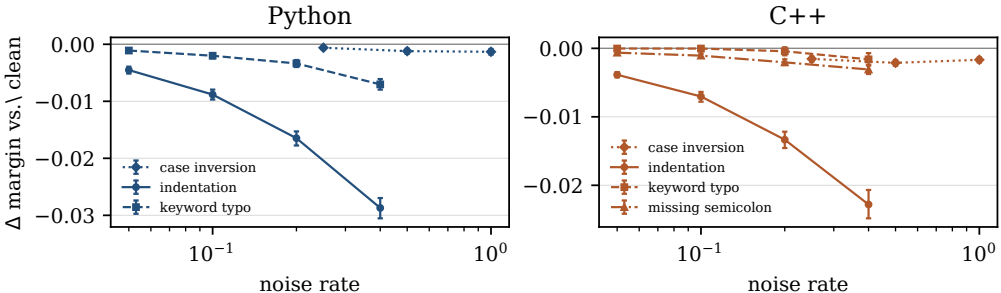


O3: depth on a common scale



- Raw tree-sitter depth counts language-specific scaffolding → count enclosing *constituents* instead, and normalise within file.
- Largest at the outermost constituents; 1 → 7 enclosing constituents: Py +0.707 → +0.519, C++ +0.553 → +0.387, En +0.714 → +0.601. Python > C++ at every depth.

O4: noise, now as a rate sweep



- Same 1,000 held-out files per language, each noise type at 5–40% (case 25–100%).
- Only indentation damage grows with rate (Py -0.005 $\rightarrow$ -0.029, C++ -0.004 $\rightarrow$ -0.023); typos cut BPP (-0.43 / -0.60 at 40%) but not alignment; renaming is flat. Every margin $\geq +0.086$.

O5: no semantic residual near memory-unsafe code

- Label audit: 12,609/12,609 regions reproduced; independent lexical labeller agrees on 433/435 statements.
- Frozen coefficient +6.93e-03 [0.0043, 0.0095] boundaries/byte — but:
- File FE only: **-0.0083 [-0.0127, -0.0039]** — unsafe statements get *fewer* boundaries (more predictable identifiers).
- Positive only after conditioning on identifier entropy; size 4.2% of the mean, 7% of a 1-SD identifier-entropy effect.
- Reading: a small conditional *surface* effect (the `*`, `&`, `new` bytes), not risk awareness.

Beyond the proposal: a 7× smaller distilled patcher

	Teacher Δ	Student Δ	kept	Student−teacher [95% CI]
Python	+0.120	+0.109	91%	-0.011 [-0.011, -0.010]
C++	+0.129	+0.119	92%	-0.011 [-0.012, -0.009]
English	+0.348	+0.304	87%	-0.044 [-0.048, -0.040]

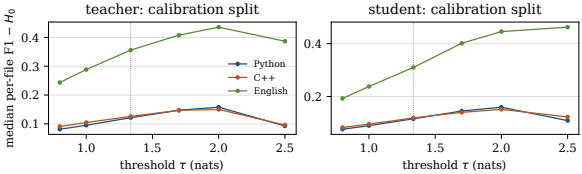
- Online KD ($\alpha = 0.1$, $T = 2$) on a disjoint 29k-file corpus; val boundary F1 vs. teacher 0.871; no-teacher control ($\alpha = 1$) 0.789.
- Full 16,300-file extraction + frozen C-v1 scoring with its own calibration ($k = 0$).
- Teacher restricted to 512-byte windows loses only +0.004 (Py) / +0.004 (C++) F1.
- Cost: forward 5.0× faster than the teacher on a RTX 2080 Ti.

Follow-ups: byte-balanced student, adapted teacher

	Balanced−student [95% CI]	Adapted−student [95% CI]	bar
Python	-0.002 [-0.003, -0.001]	+0.006 [0.004, 0.008]	0.017
C++	-0.002 [-0.0035, -0.0001]	+0.005 [0.003, 0.007]	0.022
English	+0.010 [0.008, 0.012]	+0.029 [0.026, 0.032]	0.013

- Paired Δ differences, $k = 0$, on the pre-registered 600-file code subset and the 240 English test files (exploratory: full-corpus extraction did not fit).
- Byte-balanced sampling trades code for prose: English up, code slightly down; window F1 vs. teacher 0.860 (distilled 0.871).
- Adapted teacher (99.5M, same loss and data): window F1 0.955; pre-registered verdict **not met** (1 of 3 domains beat the student's CI width).
- In code it matches the unadapted teacher read through the same windows: it recovers the teacher rather than improving on it.

Threshold sweep and the confirmatory English holdout



- Pre-registered τ^* on calibration: teacher 2; student 2/2/2.5.
- One test run at τ^*: teacher Δ +0.153 (Py), +0.158 (C++), +0.426 (En) — still below baselines (-0.210).
- Student at its own τ^*: Δ +0.147, +0.156, +0.460 — also below baselines.
- English holdout (4,107 unseen paragraphs, scored once): teacher +0.350, student +0.305; vs. word starts -0.273.

Conclusions

- Entropy boundaries track constituent starts **4–5× above chance** — held-out, holdout, and in a distilled student.
- They do **not** beat indentation / word-start segmentation at any τ or k.
- Each failed prediction has a mechanical explanation:
  - P3: margin scales with target density;
  - P4: forcing acts after the opener, not at its start;
  - O5: the "residual" is conditional, surface-level, unconditionally negative.
- Entropy is a genuine but shallow structural signal — syntax shapes it without being what it measures.